Localizing the Bottleneck in Dual-System Vision-Language-Action Models*

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***Abstract*—Dual-process architectures couple a slow semantic planner (System 2) with a fast reactive controller (System 1) for long-horizon manipulation. We evaluate a frozen Qwen2.5-VL-3B planner paired with π_0 on LIBERO-10 across six ablations (500 paired-seed episodes each). Text sub-goal injection yields no net improvement: 40.0% versus 41.0% for the no-planner baseline, 39.8% with oracle-content sub-goals on a fixed state-blind schedule (McNemar $p \geq 0.22$ everywhere), with flat steps-to-success at 151–163. Cross-conditioning 300 A1 failures on the same-seed A0 and A4 outcomes explains why: 45% fail under the bare instruction and with oracle-content sub-goals alike (S1 execution ceiling), 19% are format penalties where the appended suffix breaks a seed the bare instruction solves, 19% are seeds where oracle content recovers what Qwen's content loses, and 17% fail only under the dynamic 1 Hz stream. A same-condition noise-reseed control reshuffles an identical 39% of paired outcomes—text injection reshuffles episodes no more than the policy reshuffles itself, so every failure class but the 45% shared ceiling sits at this sampling-noise floor. The one significant per-task effect is on the most under-specified instruction, where dynamic sub-goals add $+20$ pp (p=0.013). At this controller capability level ($\sim$41%), the dominant bottleneck is S1 execution, not planner reasoning; the text interface adds variance without information, motivating learned latent bridges.**

I. Introduction

Dual-process VLA models separate a slow deliberative planner (System 2, S2) from a fast reactive controller (System 1, S1) [3]; recent instances span shared-backbone dual systems [7], learned latent bridges [5], and hierarchical transformers [6]. A shared assumption is that a better S2 improves S1. We test this directly on LIBERO-10 [8] by holding S1 fixed (π_0-finetuned, frozen) and varying S2 quality from random to oracle, with the interface constant at text sub-goal injection and no training anywhere.

Our contribution is a diagnostic ablation that addresses all three required questions via paired-seed cross-conditioning: every A1 failure is checked against a no-planner baseline (A0) and an oracle-content control (A4) on the same seed. 45% of failures persist with both controls; the rest are reshuffled at the sampling-noise floor. S1 execution is the dominant bottleneck at this capability level, not S2 reasoning.

II. Architecture and Interface

System 1 is `lerobot/pi0_libero_finetuned` [1], a 4B-parameter flow-matching policy at 25 Hz whose language encoder conditions action generation on the task string.

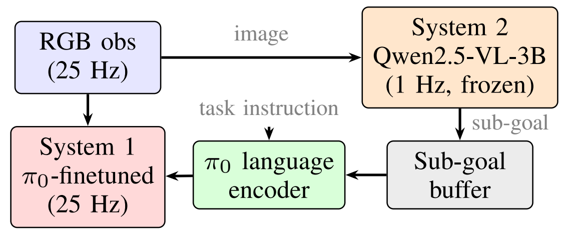

Fig. 1. Dual-system interface. S2 writes sub-goals to a shared buffer at 1 Hz; S1 reads the cached value each step and appends it to the task instruction before language encoding.

System 2 is frozen `Qwen2.5-VL-3B-Instruct` [4] (bf16, FlashAttention-2) at 1 Hz. It receives RGB + task instruction and emits a JSON sub-goal {`current_subtask`, `progress_pct`, `should_replan`}. Characterized on 866 frames: 0 parse failures; FlashAttention-2 cuts mean latency from 915 ms to 674 ms enabling stable 1 Hz operation; `progress_pct` was uniformly 0% and excluded. **S2 behavior finding:** S2 performs object-level visual grounding, not task-phase tracking (as the sub-goal string was identical across all sampled frames of a given task, e.g. always `"Grasp the alphabet soup can"` for task 0). This predicts that A1 (dynamic S2) and A2 (frozen at t=0) are functionally near-equivalent, confirmed below.

S2→S1 interface. Sub-goals inject into π_0's language encoder as **append** (`"<task>. Currently: <subtask>"`) or **replace** (sub-goal only; A3 and the A1-replace pilot). S2 runs in a background thread writing to a shared buffer that S1 reads each step, decoupling 1 Hz S2 from the 25 Hz control loop (trigger: 25 steps; Fig. 1).

III. Experimental Setup

LIBERO-10: 10 long-horizon manipulation tasks, 1000-step limit; 50 episodes per task, matching the standard 500-trial protocol [2]. Seeds are paired across all conditions; contrasts use exact McNemar tests on per-episode outcomes. No training is performed. We report SR and mean steps-to-success, measured exactly from per-step S2 traces for A1/A2/A4 and estimated from per-episode video (first 10 eps/task) for A0/A5. Oracle-content (A4) injects manually verified sub-goals (2-4

github.com/farshad-heravi/dual-system-vla-libero

*Published π_0: 85.2% on LIBERO-10 (JAX); the LeRobot PyTorch port achieves $\sim$35–41% (Issue #2114). All comparisons are internal to the port.

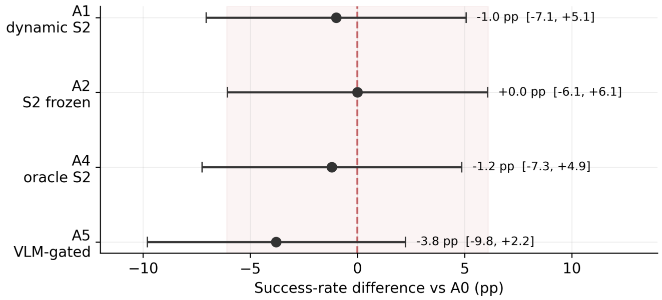

Fig. 2. Success-rate difference vs. A0 for each ablation (Newcombe 95% CI, n=500 paired seeds). The shaded band marks the ±6.1 pp same-condition noise floor from the noise-reseed control (Table II). No condition exceeds the noise floor; A3 (random replace, -41 pp) is off-scale.

ordered sub-tasks per instruction) advanced in equal time slices (state-blind, not closed-loop). Ablations are in Table I.

TABLE I
ABLATIONS ON LIBERO-10 (500 EPISODES, PAIRED SEEDS). Δ AND MCNEMAR p VS. A0. STEPS†: MEAN STEPS-TO-SUCCESS (SUCCESSFUL EPISODES ONLY).

ID	S2 source	Mode	SR (%)	Δ [p]	Steps†
A0	None	—	41.0	—	155
A2	Frozen at t=0	append	41.0	0.0 [1.00]	152
A1	Dynamic S2	append	40.0	−1.0 [0.78]	163
A4	Oracle content	append	39.8	−1.2 [0.73]	151
A5	VLM-gated	append	37.2	−3.8 [0.22]	182
A3	Random	replace	0.0	−41.0	—

IV. THE COUNTERFACTUAL: DOES SYSTEM 2 HELP?

No sub-goal condition improves over the no-S2 baseline (Table I); all $p \geq 0.22$. Steps-to-success is flat at 151–163 across A1–A4 (A0 $\approx$ A2 $\approx$ A1 $\approx$ A4 $>$ A5 $\gg$ A3). A1 (40.0%) and A2 (41.0%) are indistinguishable, as the S2 characterization predicts: dynamic updates carry no new information over a frozen-at-t=0 sub-goal when S2 never meaningfully replans. A2 is the N=∞ trigger-interval limit and matches A1 at N=25, bounding any call-rate improvement under this interface.

Reshuffling at the noise floor. A0 and A1 disagree on 209/500 seeds (107 A0-only, 102 A1-only) despite tying in aggregate. A same-condition control re-ran the dual system on 300 seeds changing *only* π_0's action-sampling noise (env seed fixed): 118/300 outcomes flip (39.3% churn, SR unmoved; Table II). The A0-vs-A1 cross-condition contrast on the same seeds also flips 118/300 (p=0.23)—identical. Swapping a real condition change for a noise reseed reshuffles no additional episodes; the text channel is a noise channel. Same-condition SR varies ∼6 pp across realizations, so sub-point deltas among A0/A1/A2/A4 carry no rank information; the ordering is a measured null, not a ranking.

Semantic coherence is critical. A3 (random, replace mode): 0%; A1-replace pilot (correct sub-goal, replace mode, 50 eps): 8.0%. π_0 actively conditions on the instruction field but does

TABLE II
SAME-CONDITION VS. CROSS-CONDITION CHURN (300 SEEDS, 30/TASK). A REAL CONDITION CHANGE RESHUFFLES EXACTLY AS MANY EPISODES AS RESEEDING ACTION-SAMPLING NOISE ALONE.

Contrast	Discordant/300	Rate	McNemar p
A1 vs. A1 (noise reseed)	118	39.3%	0.93
A0 vs. A1 (condition)	118	39.3%	0.23

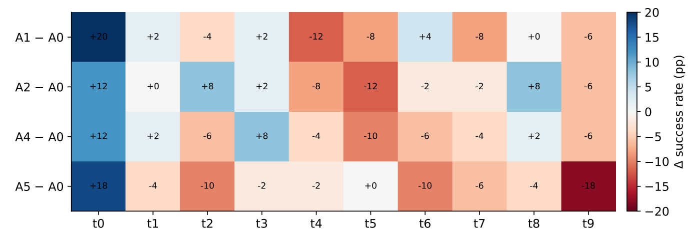

Fig. 3. Per-task success-rate change (pp) relative to A0. T0 is the only column where gains are consistent and statistically significant. All remaining cells fall within the ±6 pp same-condition noise floor established by the noise-reseed control (Table II).

not generalize to sub-goal-style text; the aggregate null is a live-channel null, not a disconnected pipeline.

One significant per-task effect. Fig. 3 maps the full per-task SR delta for every ablation. T0 ("put both the alphabet soup and the tomato sauce in the basket"): A1 lifts SR from 12% to 32% (p=0.013). T0 has the weakest baseline and most under-specified instruction; S2 supplies the missing serialization of two grasp-place cycles. Crucially, the T0 gain is column-aligned, A2 (+12 pp), A4 (+12 pp), and A5 (+18 pp) show the same direction, indicating the benefit is from instruction enrichment rather than dynamic replanning. The largest regression (T4, −12 pp, p=0.11) is not significant.

V. ERROR RECOVERY: DOES SYSTEM 2 ENABLE REPLANNING?

A5 (VLM-gated: updates only when `should_replan=true`) achieves 37.2%, the lowest and slowest instruction-bearing condition, is yet not significantly below A0 (p=0.22). The gate cannot help because S2 cannot detect scene-state changes from a single observation. To probe closed-loop recovery directly: at step 50, the task-relevant object is teleported 5 cm, forcing any pre-committed grasp to miss. Both systems run with and without the perturbation on 300 matched seeds (Table III).

Perturbation removes 13–17 pp from both systems (p<0.001); A1 does *not* retain more success under it than A0 (p=0.56). Sub-goals do change within 194/300 episodes after step 50, but the transitions track coarse task phase, not the perturbation. `should_replan` fires in none of the 300 perturbed A1 episodes: a 5 cm displacement lies below the granularity natural language can express. The text channel cannot carry the geometric correction recovery requires.

TABLE III

PERTURBATION × SYSTEM: OBJECT DISPLACED 5 CM AT STEP 50; n=300 MATCHED SEEDS PER CELL.

System	Clean	Perturbed	Δ (pp)
A0 (no S2)	34.7	21.3	−13.3
A1 (dynamic S2)	36.0	19.3	−16.7

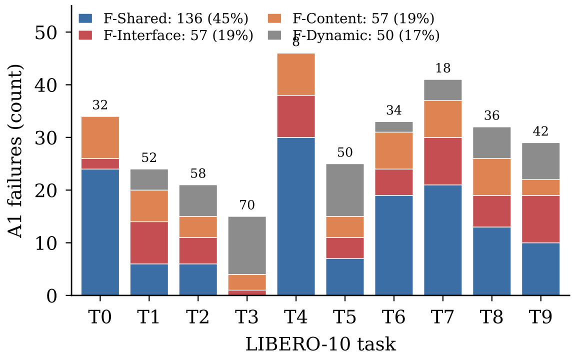

Fig. 4. Paired-seed failure partition for all 300 A1 failures by same-seed A0/A4 outcome. Numbers above bars: A1 per-task SR (%).

VI. Bottleneck Analysis

We cross-condition each of the 300 A1 failures on the A0 and A4 outcomes for the same seed. The two control bits separate four failure classes (Fig. 4):

F-Shared (136/300, 45.3%): fail under A0, A1, *and* A4 (attributable only to S1 execution). Concentrated on dual-object tasks: T4 (30/46), T0 (24/34), T7 (21/41).

F-Interface (57, 19.0%): A0 succeeds; A1 and A4 fail. Nominally the OOD suffix breaks a working episode, but this class size matches the per-direction same-condition flip count; the append-mode format penalty is not distinguishable from resampling noise. Significant costs are confined to replace mode (A3: 0%, replace pilot: 8%).

F-Content (57, 19.0%): A4 succeeds; A0 and A1 fail. Oracle content recovers what Qwen's content loses. S2-trace inspection reveals cross-task object intrusion concentrated on T4 and T8. Because A4 is state-blind, some failures may reflect timing mismatches (sub-goal 2 firing while the arm is still on sub-goal 1); F-Content is an upper bound on content-attributable failures.

F-Dynamic (50, 16.7%): both A0 and A4 succeed; only the dynamic 1 Hz stream fails. Plausible on the easiest tasks (T3, T5), but dominated by the sampling-noise floor: a pure noise reseed rescues 30% of A1 failures (58/192), while F-Content + F-Dynamic together account for 35.7%—a causal excess of only ∼6 pp (∼17 episodes) over the floor.

Verdict. Only F-Shared clears the noise floor; it dominates at 45.3%. At ∼41% controller capability, the bottleneck is execution. The per-task profile is near-frozen across all instruction-bearing runs; the lowest-baseline tasks (T0 12%, T4 20%, T7 26%) all require disambiguating two similar objects—a failure mode text injection cannot address except on T0, where serializing the two objects is itself the missing information. An interface that never broke a working episode bounds A1 at $(200{+}107)/500 = 61.4\%$, motivating learned latent bridges [5]. A degraded S1 inflates F-Shared mechanically; the same pipeline at full capability may shift the partition toward interface or content failures.

VII. Discussion

What worked. Frozen Qwen2.5-VL-3B is a reliable sub-goal generator (0/866 parse failures), and async 1 Hz execution decouples S2 from S1 without synchronization failures. Methodologically, paired-seed cross-conditioning attributes failures from existing rollouts without additional experiments, and the same-condition noise-reseed control (Table II) calibrates attribution against the policy's intrinsic sampling floor—a check any paired-seed VLA ablation should adopt.

Limitations. (i) π_0's encoder was trained only on full LIBERO task descriptions; any modification is OOD (A3: 0%; replace pilot: 8%). (ii) A4 is state-blind; sub-goal timing artifacts inflate F-Content beyond pure content failures. (iii) S2 tracks objects, not task phase (`progress_pct` uniformly 0%; `should_replan` never fires in 300 perturbed episodes). (iv) The failure partition uses a single noise-realization pair; averaging over many realizations would isolate residual causal text effects. (v) All conclusions hold at ∼41% controller capability (LeRobot port); the bottleneck ordering may shift at the 85.2% JAX reference capability.

Future work. A learned S2–S1 interface (LCB-style latent bridge [5]) removes the OOD format penalty while widening the channel; co-finetuning π_0's language adapter on auto-labeled sub-goal demonstrations is a cheaper alternative. A temporally conditioned S2 (observation–outcome history) could enable genuine phase-transition and perturbation detection; re-running the A5 VLM-gated condition with such a memory-augmented S2 would test whether the `should_replan` gate can fire on perturbations that a single-frame S2 misses.

References

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